Perspective-Conditioned Geopolitical Event Prediction

A Full Architecture Specification

Abstract. This document specifies a neural architecture designed to predict whether one real-world entity will take a meaningful action toward another. The model runs N independent sub-models, one per personality perspective, and aggregates their votes. Every prediction is conditioned on N global context vectors describing the current state of the world. Embeddings are produced by a trained MiniLM converter that maps natural language strings into the model's custom 64-dimensional spaces. The final output is a two-number probability vector. N is left open pending experimental design.

Embedding Pipeline and the MiniLM Converter

Before any forward pass through the architecture, each input string must be converted into a dense 64-dimensional vector in the appropriate embedding space. This is handled by three independent **converter models**, one per space.

Why a converter?

MiniLM is a pretrained sentence transformer that produces 384-dimensional embeddings from arbitrary natural language strings. It has already been trained on vast amounts of text and understands semantic similarity out of the box. However, our architecture requires 64-dimensional vectors in three *distinct* spaces (entity, personality, context) that have been shaped by triplet training to reflect geopolitical similarity structure. A naive linear projection from MiniLM's 384-d space to 64-d would discard most of the semantic structure.

Instead, each converter is a small learned network trained to map MiniLM's output into the corresponding custom space:

$$\text{converter}_k : \mathbb{R}^{384} \longrightarrow \mathbb{R}^{64} \qquad k \in \{\,\text{entity},\ \text{personality},\ \text{context}\,\}$$

Each converter consists of a two-layer MLP with a LayerNorm output, trained with triplet loss so that its 64-d output space preserves the similarity structure defined by the training triplets.

Full embedding pipeline for a string

Given any natural language string s (e.g. "United States", "military analyst", "ongoing civil war in Sudan 2003"):

$$s \xrightarrow{\text{MiniLM}} \mathbf{z} \in \mathbb{R}^{384} \xrightarrow{\text{converter}_k} \mathbf{e} \in \mathbb{R}^{64}$$

This two-step pipeline means the architecture can embed *any string it has never seen before* at inference time, not just strings present in the training vocabulary. New entities, new personalities, and new context descriptions can be fed in directly without retraining the embedding tables.

The three converters

Entity converter maps actor and receiver strings into the shared 64-d entity space. Both A and B are produced by the same converter from the same weights, which is what makes their cosine similarity meaningful.

Personality converter maps personality description strings into the 64-d personality space. Examples: "realist international relations scholar", "pacifist humanitarian analyst".

Context converter maps natural language context strings into the 64-d context space. These strings are assembled from COW datasets (National Material Capabilities, MIDs, Formal Alliances, War Data) and describe the global situation at a given point in time. Example: "Year 1990. Active conflict: Gulf War beginning, Iraq invades Kuwait. US dominant CINC score. No alliance between actors."

The Four Inputs

After the converter pipeline, every forward pass receives:

- $A \in \mathbb{R}^{64}$ the **actor** embedding
- $B \in \mathbb{R}^{64}$ the **receiver** embedding (same converter and space as A)
- $\{C_i\}_{i=0}^{N-1},\ C_i \in \mathbb{R}^{64}$ bank of N personality embeddings

- $\{D_j\}_{j=0}^{N-1},\ D_j \in \mathbb{R}^{64}$ bank of N context embeddings

All four use $d = p = q = 64$ in this implementation. A and B share one converter; C and D each have their own. No weights are shared across the three converter families.

Stage 1 The Personality–Context Matrix

Every personality is paired with every context via an outer product, producing a matrix $\mathbf{M}$ that holds all N^2 combinations:

$$\mathbf{M}_{ij} \;=\; C_i \otimes D_j \;\in\; \mathbb{R}^{64\times 64}$$

Each cell is a 64×64 matrix (4096 values). The full tensor is:

$$\mathbf{M} \;\in\; \mathbb{R}^{N\times N\times 64\times 64}$$

which we flatten the last two dimensions of to get:

$$\mathbf{M} \;\in\; \mathbb{R}^{N\times N\times 4096}$$

Row i contains personality i paired with every context. Column j contains context j seen through every personality.

Data shape after Stage 1: $\mathbf{M} \in \mathbb{R}^{B\times N\times N\times 4096}$ where B is the batch size.

Stage 2 Residual Refinement

Three sequential residual passes refine $\mathbf{M}$. Starting from $\mathbf{M}^{(0)} = \mathbf{M}$:

$$\mathbf{M}^{(t)} \;=\; (1+\alpha)\cdot \mathbf{M}^{(t-1)} \;+\; \mathbf{B} \qquad t = 1,\, 2,\, 3$$

$\alpha \in \mathbb{R}$ is a single shared scalar learned across all cells and all passes. $\mathbf{B} \in \mathbb{R}^{N\times N\times 4096}$ is a per-cell learned bias — each of the N^2 cells has its own 4096-dimensional bias vector. The $(1+\alpha)$ factor is the scaled copy; the original matrix is preserved inside the $(1+\alpha)$ multiplication because the scale includes the identity contribution.

Expanding all three passes:

$$\mathbf{M}^{(1)} = (1+\alpha)\,\mathbf{M} + \mathbf{B}$$
$$\mathbf{M}^{(2)} = (1+\alpha)\,\mathbf{M}^{(1)} + \mathbf{B}$$
$$\mathbf{M}^{(3)} = (1+\alpha)\,\mathbf{M}^{(2)} + \mathbf{B} \quad \equiv \quad \mathbf{M}'$$

Data shape after Stage 2: $\mathbf{M}' \in \mathbb{R}^{B\times N\times N\times 4096}$ — same shape as $\mathbf{M}$, values refined.

Stage 3 Projection into Actor–Target Space

$\mathbf{M}'$ lives in personality–context space (4096-d per cell). To interact with A and B (64-d), each cell must be projected down to 64-d. Two learned linear layers W_Q and W_K handle this, applied identically to every cell:

$$W_Q,\, W_K \;:\; \mathbb{R}^{4096} \;\longrightarrow\; \mathbb{R}^{64}$$

Applied across all cells:

$$\mathbf{Q} = W_Q(\mathbf{M}') \;\in\; \mathbb{R}^{B\times N\times N\times 64}$$
$$\mathbf{K} = W_K(\mathbf{M}') \;\in\; \mathbb{R}^{B\times N\times N\times 64}$$

We then average over the context axis (axis 2, the N columns) to collapse each personality's N context projections into a single 64-d vector:

$$\bar{\mathbf{Q}} = \mathrm{mean}_j(\mathbf{Q}) \;\in\; \mathbb{R}^{B\times N\times 64}$$
$$\bar{\mathbf{K}} = \mathrm{mean}_j(\mathbf{K}) \;\in\; \mathbb{R}^{B\times N\times 64}$$

Data shape after Stage 3: $\bar{\mathbf{Q}},\, \bar{\mathbf{K}} \in \mathbb{R}^{B\times N\times 64}$ — one 64-d vector per personality, in actor–target space.

Stage 4 Blending Actor and Receiver

Cosine similarity

$$\cos_{AB} \;=\; \frac{A\cdot B}{\|A\|\,\|B\|} \;\in\; [-1,\,1]$$

Near $+1$: A and B are similar (allies, ideological partners). Near 0: unrelated. Near -1: opposites or adversaries.

The blended vector

$$\mathrm{AB}_n \;=\; \tanh\big(A \cdot (1 - \cos_{AB})\big) \;+\; \tanh\big(B \cdot \cos_{AB}\big) \;\in\; \mathbb{R}^{64}$$

When A and B are dissimilar, the left term dominates and the blend is driven by the actor. When they are similar, the right term dominates and the receiver's character shapes the blend. Tanh on each term keeps both components bounded before summing.

Context fusion

AB_n is fused with each context vector D_j through a per-context learned element-wise transform:

$$\mathrm{ABD}_j \;=\; \tanh\big(w_j \;\odot\; \mathrm{AB}_n \;\odot\; D'_j \;+\; b_j\big) \;\in\; \mathbb{R}^{64}$$

where $D'_j = \mathrm{Linear}_{q\to d}(D_j)$ is D_j projected to 64-d, and w_j, $b_j \in \mathbb{R}^{64}$ are learned per-context parameters.

Data shape after Stage 4: $\mathrm{ABD}_j \in \mathbb{R}^{B\times N\times 64}$ — one fused vector per context j.

Stage 5 The N Mini Models

N independent sub-models run in parallel, one per personality. Each mini-model produces: (1) a refined personality vector, (2) a two-number probability vote conditioned on both personality and situation, and (3) a confidence-gated output vector.

Step A: normalise

$$\tilde{C}_i \;=\; \mathrm{softmax}(N \cdot C_i) \;\in\; \mathbb{R}^{64}$$

Multiplying by N before softmax sharpens the distribution so dominant personality dimensions stand out.

Step B: affine + ReLU

$$C^*_i \;=\; \mathrm{ReLU}\big(W^{(1)}_i\, \tilde{C}_i + b^{(1)}_i\big) \;\in\; \mathbb{R}^{64}$$

$W_i^{(1)} \in \mathbb{R}^{64\times 64}$ and $b_i^{(1)} \in \mathbb{R}^{64}$ are *unique to mini-model i* — no two mini-models share these weights.

Step C: probability vote

The probability head concatenates the personality representation C_i^* with the situation vector $\mathrm{ABD}_j[:, i, :]$ so that each vote is conditioned on both the personality *and* the current actor–receiver–context:

$$\hat{p}_i \;=\; \mathrm{softmax}\Big(W_i^{(\mathrm{prob})}\,\big[C_i^* \parallel \mathrm{ABD}_i\big] + b_i^{(\mathrm{prob})}\Big) \;\in\; \mathbb{R}^2$$

where $W_i^{(\mathrm{prob})} \in \mathbb{R}^{2\times 128}$, $\big[\cdot\|\cdot\big]$ is concatenation, and $\hat{p}_i = [\hat{p}_i^+,\, \hat{p}_i^-]$ with $\hat{p}_i^+ + \hat{p}_i^- = 1$.

Step D: confidence gating

$$C_i'' \;=\; \tanh\big(\hat{p}_i^+\, C_i^*\big) \;+\; \tanh\big(\hat{p}_i^-\, C_i^*\big) \;\in\; \mathbb{R}^{64}$$

A confident mini-model (one of $\hat{p}_i^{\pm}$ near 1) pushes C_i'' toward tanh saturation, producing a strong contribution. An uncertain mini-model (both near 0.5) produces a soft, moderate C_i''.

Ensemble matrix

$$\mathbf{E} \;=\; \begin{pmatrix} \longleftarrow C_0'' \longrightarrow \\ \longleftarrow C_1'' \longrightarrow \\ \vdots \\ \longleftarrow C_{N-1}'' \longrightarrow \end{pmatrix} \;\in\; \mathbb{R}^{N\times 64}$$

Data shape after Stage 5: $\mathbf{E} \in \mathbb{R}^{B\times N\times 64}$, $\hat{p} \in \mathbb{R}^{B\times N\times 2}$

Stage 6 Polling the Ensemble

The fused situation vector ABD_j is dotted against $\mathbf{E}$ to score every personality's relevance to the current situation. We use the mean across the N context slots of ABD as the query:

$$\bar{\mathrm{ABD}} \;=\; \mathrm{mean}_j(\mathrm{ABD}_j) \;\in\; \mathbb{R}^{B\times 64}$$

$$v \;=\; \bar{\mathrm{ABD}} \cdot \mathbf{E}^\top \;\in\; \mathbb{R}^{B\times N}$$

Each element v_i measures alignment between the current actor–receiver–context signal and personality i's gated view.

Data shape after Stage 6: $v \in \mathbb{R}^{B \times N}$

Stage 7 Bottleneck Refinement

Three expand–contract passes compress and re-expand v, forcing the model to find a compact summary of all N votes:

$$v^{(t)} \;=\; W_{\downarrow}^{(t)}\,\mathrm{ReLU}\big(W_{\uparrow}^{(t)}\,v^{(t-1)}\big)$$

$$W_{\uparrow}^{(t)} \in \mathbb{R}^{128 \times N} \quad \text{(expand)}, \qquad W_{\downarrow}^{(t)} \in \mathbb{R}^{N \times 128} \quad \text{(contract)}$$

$$v^{(t-1)} \in \mathbb{R}^{N} \xrightarrow{W_{\uparrow}} \mathbb{R}^{128} \xrightarrow{\mathrm{ReLU}} \mathbb{R}^{128} \xrightarrow{W_{\downarrow}} v^{(t)} \in \mathbb{R}^{N}$$

Applied three times from $v^{(0)} = v$, yielding $v^{(3)}$. Each pass has its own independent $W_{\uparrow}^{(t)}$ and $W_{\downarrow}^{(t)}$.

Data shape after Stage 7: $v^{(3)} \in \mathbb{R}^{B \times N}$

Stage 8 Final Output

$$\hat{y} \;=\; \mathrm{softmax}\big(W_{\mathrm{out}}\,v^{(3)}\big) \;\in\; \Delta^{1}, \qquad \hat{y} = [\hat{y}^{+},\ \hat{y}^{-}], \quad \hat{y}^{+} + \hat{y}^{-} = 1$$

$W_{\mathrm{out}} \in \mathbb{R}^{2 \times N}$ is a learned output projection. $\hat{y}^{+}$ is the probability entity A takes the action toward B; $\hat{y}^{-}$ is the probability it does not.

End-to-End Data Flow

Embedding pipeline (pre-forward):

$$\text{string} \xrightarrow{\text{MiniLM}} \mathbb{R}^{384} \xrightarrow{\text{converter}_k} \mathbb{R}^{64}$$

Forward pass:

$$(A,\, B,\, \{C_i\},\, \{D_j\}) \;\in\; \mathbb{R}^{64} \;\longrightarrow\; \mathbf{M} \in \mathbb{R}^{B\times N\times N\times 4096}$$

$$\xrightarrow{\text{3 residual passes}} \mathbf{M}' \in \mathbb{R}^{B\times N\times N\times 4096}$$

$$\xrightarrow{W_Q, W_K\,:\,4096\to 64} \bar{\mathbf{Q}},\, \bar{\mathbf{K}} \in \mathbb{R}^{B\times N\times 64}$$

$$\xrightarrow{\cos_{AB},\,\text{blend}} \mathrm{AB}_n \in \mathbb{R}^{B\times 64} \xrightarrow{\odot D_j} \mathrm{ABD}_j \in \mathbb{R}^{B\times N\times 64}$$

$$\xrightarrow{N\ \text{mini-models}} \mathbf{E} \in \mathbb{R}^{B\times N\times 64}, \quad \hat{p} \in \mathbb{R}^{B\times N\times 2}$$

$$\xrightarrow{\cdot\,\mathbf{E}^\top} v \in \mathbb{R}^{B\times N} \xrightarrow{\text{3 bottleneck passes}} v^{(3)} \in \mathbb{R}^{B\times N} \xrightarrow{W_{\text{out}}} \hat{y} \in \Delta^1$$

Notation Reference

Symbol	Meaning
A, $B \in \mathbb{R}^{64}$	Actor and receiver embeddings, shared space
$C_i \in \mathbb{R}^{64}$	i-th personality embedding
$D_j \in \mathbb{R}^{64}$	j-th context embedding
N	Number of personalities / contexts (TBD)
$\mathbf{M}$	$N \times N \times 4096$ outer-product tensor
$\mathbf{M}'$	$\mathbf{M}$ after 3 residual passes
α	Shared scalar residual weight
$\mathbf{B}$	Per-cell bias, $\mathbb{R}^{N \times N \times 4096}$
W_Q, W_K	Linear projections $4096 \to 64$
$\bar{\mathbf{Q}}$, $\bar{\mathbf{K}}$	Context-averaged projections, $\mathbb{R}^{B \times N \times 64}$
$\cos_{AB}$	Cosine similarity of A and B
AB_n	Blended actor–receiver vector, $\mathbb{R}^{64}$
ABD_j	Context-fused vector, $\mathbb{R}^{64}$
$\hat{p}_i \in \mathbb{R}^2$	Mini-model i probability vote
C_i^*	Post-ReLU personality, $\mathbb{R}^{64}$
C_i''	Confidence-gated personality, $\mathbb{R}^{64}$
$\mathbf{E} \in \mathbb{R}^{N \times 64}$	Ensemble matrix
$v \in \mathbb{R}^N$	Raw poll vector
$v^{(3)} \in \mathbb{R}^N$	Refined poll vector
$\hat{y} \in \Delta^1$	Final binary probability
$\mathrm{converter}_k$	MiniLM $\to$ 64-d learned projection, space k
$\odot$	Element-wise product
$\otimes$	Outer product
$[\cdot \| \cdot]$	Vector concatenation

Note: N is an unspecified hyperparameter controlling the size of the personality bank, context bank, cross-product matrix, and ensemble. It will be fixed during experimental design.